

Answer Key & Teacher Notes

Session 1 – Substitute and Solve • Session 2 – Working Backward to a Goal

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	First 5–8 minutes of Session 1. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 • TEACHER NOTES

Substitute and Solve

TALK PROMPTS

- Why should you check an answer against the situation before you trust it?
- How would you check that 216 candles makes sense for 12 hours of work?

PARTNER TASK

Muffin Trays. A bakery bakes 12 muffins on every tray. One partner writes $m = 12t$ and finds the muffins on 14 trays (168); the other estimates ($10 \times 15 = 150$) to check that the answer is reasonable. Switch with Keisha rowing 40 meters per minute for 12 minutes (480).

WATCH FOR

- Adding the input to the rate instead of multiplying (6m with $m = 30$ gives 36).
- Adding the fee before multiplying ($(14 + 25) \times 6$ instead of $14 \times 6 + 25$).
- Giving a number without units, or not answering the yes/no question.

SESSION 2 • TEACHER NOTES

Working Backward to a Goal

TALK PROMPTS

- Which is easier for the class to control: the price or the number sold? Why?
- How would your work change if the club raised the ticket price but kept the same goal?

PARTNER TASK

Candle Sale Two Ways. A club needs \$540. One partner finds the candles needed at \$9 each (60); the other finds the price if the club sells 45 candles (\$12). Then each checks the other by multiplying.

WATCH FOR

- Multiplying by the rate instead of dividing ($4b = 96$ answered as 384).
- Dividing before subtracting the fee ($600 \div 15 = 40$ instead of $510 \div 15 = 34$).
- Rounding 37.5 tickets down to 37, which does not reach the goal.

ITEM ANALYSIS

depth of knowledge (DOK) for each practice item

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 • Form A	6.EE.C.9	1	2	1	1	2	2	1	3	3	1	1
S1 • Form B	6.EE.C.9	1	2	1	1	2	2	1	3	3	1	1
S2 • Form A	6.EE.C.9	1	2	1	1	2	3	2	3	3	1	1
S2 • Form B	6.EE.C.9	1	2	1	1	2	3	2	3	3	1	1

Items before the last three are Practice; then Find the Error (DOK 3) and two Test Practice items.

GUIDED NOTES

substitute: Put a known number in **place** of a letter.

order of operations: **Multiply** first, then add or subtract.

reasonable: Makes **sense** for the story.

- To find an output, **substitute** the input for the variable.
- With a fee, **multiply** first, then add the fee.

Example: Substitute $h = 5$: $r = 24 \times 5$. Multiply: $r = 120$ rolls. Compare: $120 < 150$. Answer: **No**, 5 hours makes only 120 rolls.

YOU TRY

1. a. 31 b. 108
2. $8 \times 5 + 15 = 40 + 15 = \55

EXIT TICKET

1. a. 60 b. 23
2. $14 \times 35 = 490$ boxes
3. **B** (22)

PRACTICE FORM A

1. a. 72 b. 30 c. 47 d. 73
2. a. $y = 18x$ b. $18 \times 12 = 216$ candles c. No: $18 \times 7 = 126$, which is less than 900
3. $c = 24t$; $24 \times 16 = 384$ cookies
4. $m = 50n$; $50 \times 9 = 450$ meters
5. Yes. $8 \times 45 = 360$ sandbags, which matches the claim.
6. $b = 2.5h$; $2.5 \times 24 = 60$ boards
7. $14 \times 6 + 25 = 84 + 25 = \109
8. a. $3 \times 15 + 20 = \$65$ b. $3 \times 30 + 20 = \$110$ c. \$110 is not $2 \times \$65 = \130 , because the \$20 fee is paid only once.
9. He added $6 + 30$ instead of multiplying. 6m means $6 \times m$, so $b = 6 \times 30 = 180$ buckets.
10. **B** (48)
11. **B** (multiply first, then add the fee)

Self-check word **ANSWER**

PRACTICE FORM B

1. a. 56 b. 32 c. 59 d. 68
2. a. $y = 16x$ b. $16 \times 11 = 176$ bars c. Yes: $16 \times 8 = 128$, which is at least 120
3. $m = 12t$; $12 \times 14 = 168$ muffins
4. $d = 40n$; $40 \times 12 = 480$ meters
5. No. $9 \times 30 = 270$ boxes, not 290, so the claim does not fit.
6. $b = 1.5h$; $1.5 \times 16 = 24$ bikes
7. $12 \times 5 + 30 = 60 + 30 = \90
8. a. $4 \times 10 + 15 = \$55$ b. $4 \times 20 + 15 = \$95$ c. \$95 is not $2 \times \$55 = \110 , because the \$15 fee is paid only once.
9. He added $8 + 20$ instead of multiplying. 8m means $8 \times m$, so $j = 8 \times 20 = 160$ jugs.
10. **B** (63)
11. **B** (multiply 7 by 3)

Self-check word **RESULT**

GUIDED NOTES

work backward: Start with the **goal** and undo the steps.

inverse operation: An action that **undoes** another action.

at least: That number **or more**.

- Put the goal in place of the **output**, then solve for the input.
- With a fee, **subtract** the fee first, then divide by the rate.
- If only whole items make sense, round **up** to reach the goal.

Example: Put in the goal: $100 = 6 \cdot n$. Divide: $100 \div 6 \approx 16.7$. 16 tickets make \$96 (too low); 17* tickets make \$102. Answer: The club must sell **17** tickets.

YOU TRY

- a. $x = 48$ b. $x = 10$
- $60 \div 8 = 7.5$; 7 vans hold only 56, so 8 vans

EXIT TICKET

- a. $n = 12$ b. $n = 11$
- $11c = 528$; $c = 528 \div 11 = 48$ cars
- B** (subtracting 6)

PRACTICE FORM A

- a. $x = 14$ b. $x = 15$ c. $x = 11$ d. $x = 25$
- a. $m = 12p$ b. $660 = 12p$, so $p = 660 \div 12 = 55$ pots
c. $660 \div 44 = \$15$ each
- $3n = 210$; $n = 210 \div 3 = 70$ pages
- $7t = 350$; $t = 350 \div 7 = 50$ tickets
- $r = 9m$; $135 = 9m$, so $m = 135 \div 9 = 15$ minutes
- No: $13 \times 25 = \$325$, which is more than \$310. Most:
 $310 \div 25 \approx 12.4$, so 12 tickets.
- a. $m = 15k + 90$ b. $600 - 90 = 510$; $510 \div 15 = 34$ tickets
- a. $t = 300 \div 8 = 37.5$ b. 38 tickets ($37 \times 8 = \$296$ is not enough; $38 \times 8 = \$304$)
- She multiplied 96×4 . To undo multiplying by 4, divide: $b = 96 \div 4 = 24$ wristbands. Check: $4 \times 24 = 96$.
- B** (13)
- C** (8)

Self-check word INVERSE

PRACTICE FORM B

- a. $x = 12$ b. $x = 17$ c. $x = 13$ d. $x = 15$
- a. $m = 9p$ b. $540 = 9p$, so $p = 540 \div 9 = 60$ candles
c. $540 \div 45 = \$12$ each
- $5n = 185$; $n = 185 \div 5 = 37$ laps
- $6t = 324$; $t = 324 \div 6 = 54$ tickets
- $r = 7m$; $168 = 7m$, so $m = 168 \div 7 = 24$ minutes
- Yes: $11 \times 18 = \$198$, which is within \$200. Most: $200 \div 18 \approx 11.1$, so 11 tickets.
- a. $m = 16k + 80$ b. $560 - 80 = 480$; $480 \div 16 = 30$ T-shirts
- a. $t = 300 \div 9 = 33.3\ldots$ b. 34 tickets ($33 \times 9 = \$297$ is not enough; $34 \times 9 = \$306$)
- He multiplied 85×5 . To undo multiplying by 5, divide: $s = 85 \div 5 = 17$ stickers. Check: $5 \times 17 = 85$.
- A** (17)
- B** (6)

Self-check word REVERSE

HOMEWORK PACKET • ANSWERS

same answers for the core packet and Level 1

SHEET 1 • SUBSTITUTE AND SOLVE

1. a. 19 b. 72 c. 28 d. 42
2. 3, 5, 7, 11, 13; points (0, 3), (1, 5), (2, 7), (4, 11), (5, 13) on a straight line
3. a. $15 \times 6 + 20 = 110$ gal b. $15 \times 10 + 20 = 170$ gal
4. 3 hours: $7 \times 3 + 5 = \$26$; 8 hours: $7 \times 8 + 5 = \$61$
5. a. Yes: $5 \times 4 + 2 = 22$ b. No: $5 \times 6 + 2 = 32$, not 30
6. He added before multiplying. Multiply first: $4 \times 5 = 20$, then $20 + 3 = 23$. $y = 23$.

SHEET 2 • WORKING BACKWARD TO A GOAL

1. a. 9 b. 13 c. 9 d. 8
2. $210 - 45 = 165$; $165 \div 15 = 11$ weeks
3. $150 - 30 = 120$; $120 \div 12 = 10$ weeks. Check: $12 \times 10 + 30 = 150$.
4. a. week 4 b. $33 - 6 = 27$; $27 \div 3 = 9$, so week 9
5. $150 \div 12 = 12.5$. She cannot buy half a box, and 12 boxes (144) is not enough, so she buys 13 boxes.
6. **B** (6 hours)
7. **C** ($x = 13$)